AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 21 · STUDENT LESSON

Caching and stale information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Check permission. Match context. Check age.

Explain why a young saved answer can still be wrong for the current request.

Keep one habit: Complete contract → main decision ledger → versioned opening-time picture → boundary probes → local cache audit → independent version change.

MAIN LESSON · SECTION 1

Reuse an answer only while its context still matches

> A saved answer needs current permission, matching versions and a valid age.

Learning goals

I can explain the **mechanism** and terminology of caching and stale information.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve access and retention distinctions

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Data retention and deletion. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain caching and stale information using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of caching and stale information

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Separate permission, matching context and freshness

> Passing one check does not excuse a failure in another.

Begin with a saved opening-time note

A cache is a saved result that can avoid repeating work. Think of keeping yesterday’s opening-time note: it is useful only if you are allowed to read it and it still belongs to the approved information. A young entry can still be wrong for this request. Check current permission first, then match the query, trusted role, source collection version and procedure version, then check the entry’s age.

Define the mechanism

Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries.

Complete access-aware cache contract

Check	Complete classroom rule
Current permission	Trusted Boolean fixture checked before any cached value is returned
Cache key	Query, trusted role, corpus version and procedure version

Check	Complete classroom rule
Trusted role	Student or teacher supplied by the harness; not a request-body claim
Age and TTL units	Virtual minutes; age = now minus saved time
Freshness	0 ≤ age < TTL; exact equality expires
Missing or changed context	Miss: do not return a differently keyed entry
Miss is not an answer	Only an authorized lookup of approved evidence can supply a replacement
Scope	Role-only separation is sufficient only for these fixtures; real user, tenant or object differences need appropriate additional scope
What TTL cannot prove	Truth, current permission, correct publication versions or physical deletion

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Trace a young entry that still must not be served

> Every expected result states which check decides it.

Original four-part key and strict age rule

Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

Complete main fixture decision ledger

Main fixture: TTL 5 minutes	Request at virtual minute	Expected result and reason
Teacher c1/p1 entry saved at 0; teacher c1/p1 allowed	4	Hit; matching key and age 4 < 5
Same query, student c1/p1 allowed	4	Miss; teacher entry has a different role key
Teacher c1/p1 permission revoked	4	Denied before cached content is returned
Teacher c1/p1 allowed	5	Expired miss; age equals TTL
Teacher c2/p1 allowed	1	Miss; corpus version changed despite young age
Teacher c1/p2 allowed	1	Miss; procedure version changed despite young age

State time units and current context

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Caching and stale information — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Teacher response is cached by query only and returned to a student.	Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.	Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict before revealing a matching-version cache hit

> Hold role and procedure fixed while changing the source-version match.

Current task and response

- Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the next approved-version request

- Change: The next request still uses approved version 5.

▼ Check after predicting

Prediction: The matching version 5 entry can supply 09:30 under the freshness policy.

Reason: Request and cached source version now agree.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read the old-version miss and the later fresh matching hit

> A version match is necessary but does not replace current permission or freshness.

Main query-only cache leak and repair

Teacher response is cached by query only and returned to a student.

Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate opening-time inputs and complete context

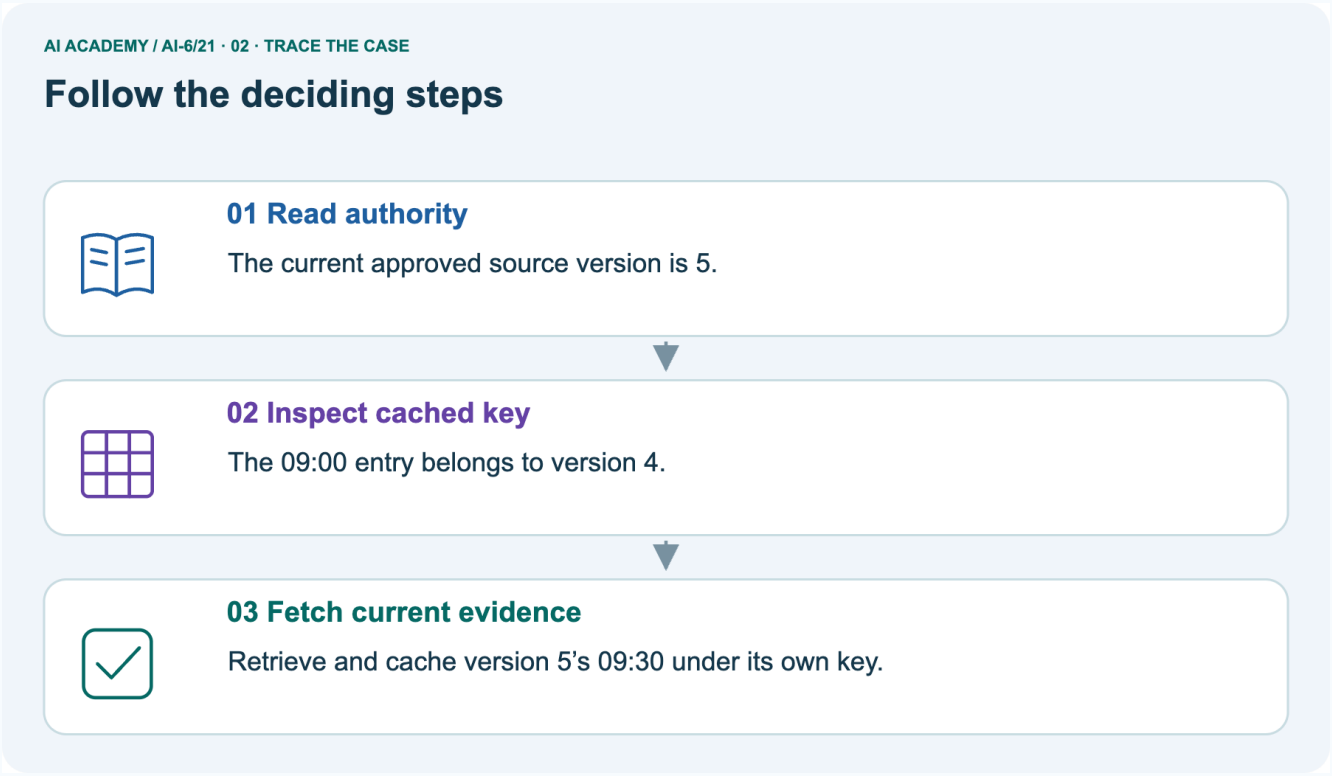
Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

The picture simplifies the key to room and source version while holding the trusted role and procedure fixed. Approved version 5 says 09:30; the saved version 4 entry says 09:00. The first version 5 request must miss that old key even if the entry is young. Its authorized local fixture lookup supplies 09:30 and saves a separate version 5 entry.

A later request may hit that entry only while permission still applies and its age remains below TTL. Reliable publication must change the version when evidence changes; a silent edit under the same version is invisible to these key and age checks.

Trace approved version 5 rather than saved version 4



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read authority	The current approved source version is 5.
2. Inspect cached key	The 09:00 entry belongs to version 4.
3. Fetch current evidence	Retrieve and cache version 5's 09:30 under its own key.

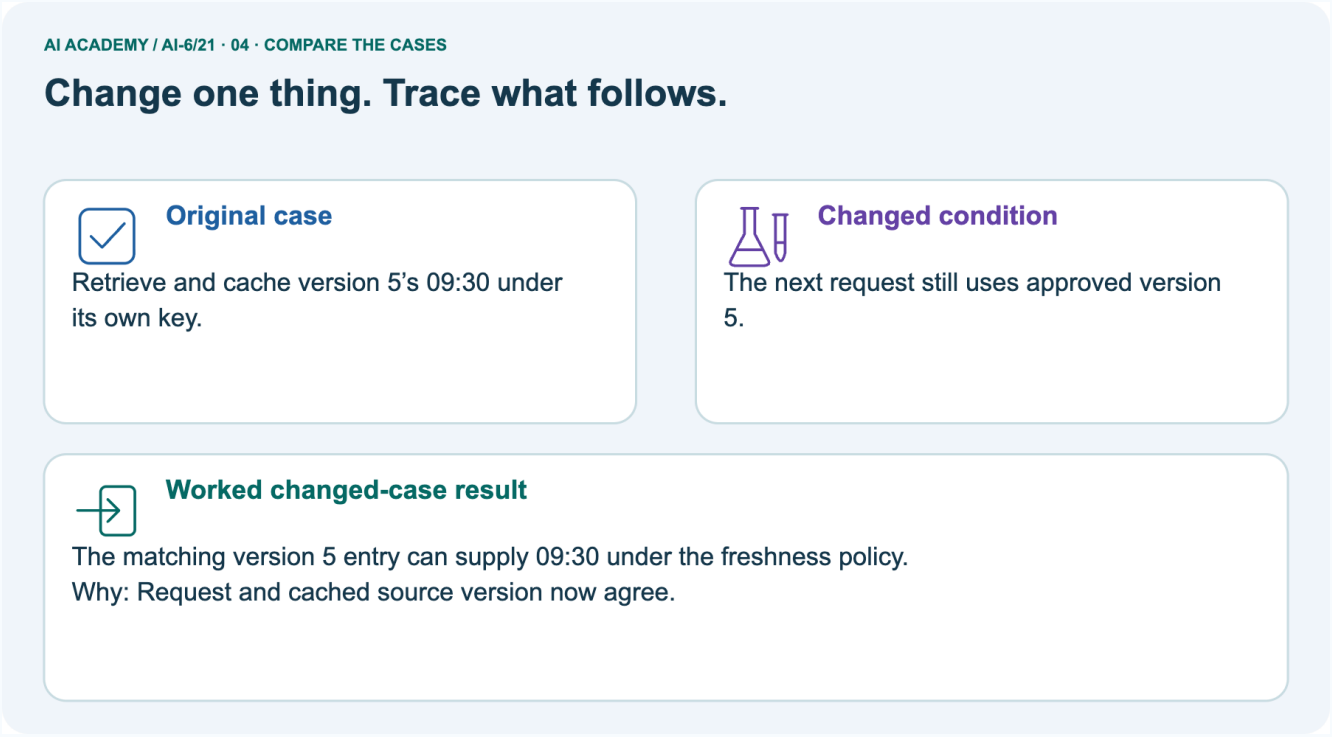
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

Predict the next matching-version request

- Change one thing: The next request still uses approved version 5. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain silent-edit limits and first/later requests

- Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

- Read the labelled picture: Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?

▼ Earlier wording, alternatives and picture transcripts

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def cache_key(query, role, corpus_version, procedure_version):
    return (query, role, corpus_version, procedure_version)

def cache_valid(age, ttl, saved_version, current_version):
    return 0 <= age < ttl and saved_version == current_version

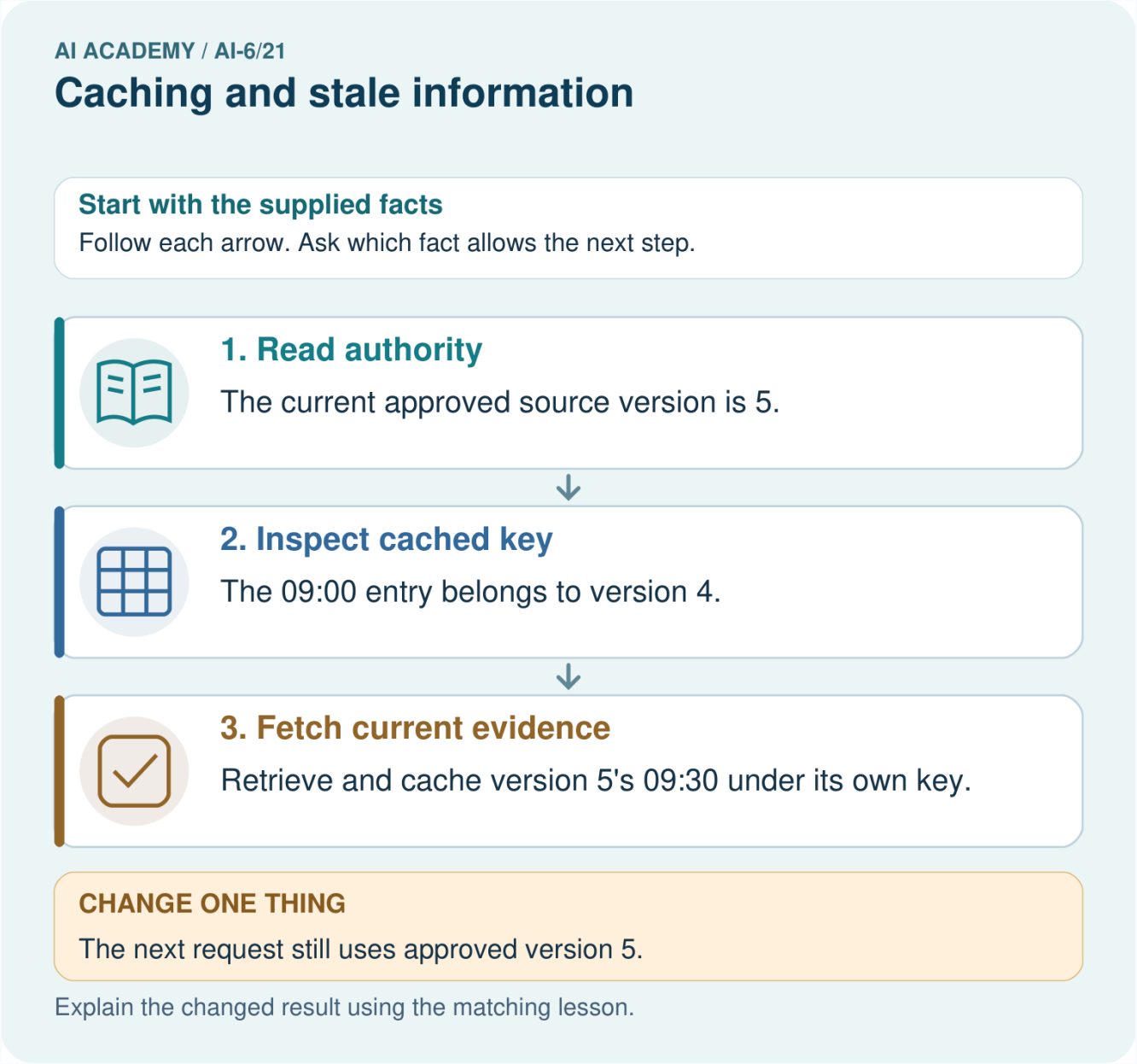
print(cache_valid(10, 10, "v1", "v1"))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Caching and stale information

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit stale information, access leaks and overclaims separately

> Cache eligibility does not establish correctness or deletion.

Core qualification and failure analysis

A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete context, boundary and publication probes

Probe	Expected interpretation
Age just below TTL with matching context and permission	May serve the saved authorized value; no new truth guarantee
Age exactly TTL or above	Miss; expired
Negative age	Miss; inconsistent supplied timestamps
NaN time or nonpositive TTL	Reject invalid fixture
Permission changes after saving	Denied even while entry is fresh
Role changes but query stays the same	Different key; never reuse the teacher-only value for a student
Corpus or procedure version changes	Different key; young old-version entry cannot satisfy it

Probe	Expected interpretation
Source changes without changing version	May go undetected; publication process must maintain versions
Expired entry still exists in dictionary	Not served, but not deleted; retention is a separate policy

Guided repair and explanation

- A report omits this qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise one changed condition at a time

> Say which check fails first and what evidence remains unavailable.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example “Caching and stale information”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Caching and stale information. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?

My explanation using a pictured step and the given facts

Reserve the selected version-change case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete local cache decision investigation

> Inspect decision statuses and the approved source fixture lookup count.

Current query-only leak investigation

Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

Runnable permission/version/age model

```
"""Local access-aware/versioned cache fixtures with virtual minute timestamps."""
from copy import deepcopy
from math import isfinite

def finite_nonnegative(value):
    return type(value) in (int,float) and isfinite(value) and value>=0

def cache_key(query,role,corpus_version,procedure_version):
    if any(type(v) is not str or not v for v in
[query,role,corpus_version,procedure_version]):
        raise ValueError('nonempty key fields required')
    if role not in ('student','teacher'):raise ValueError('unknown trusted role fixture')
    return query,role,corpus_version,procedure_version

class FixtureCache:
    def __init__(self):self.entries={}
    def put(self,query,role,corpus_version,procedure_version,value,saved_at):
        if not finite_nonnegative(saved_at):raise ValueError('invalid virtual save time')
        key=cache_key(query,role,corpus_version,procedure_version)
        self.entries[key]={'saved_at':saved_at,'value':deepcopy(value)}
    def get(self,query,role,corpus_version,procedure_version,now,ttl,*,currently_allowed):
        if type(currently_allowed) is not bool:raise ValueError('trusted permission
fixture required')
        if not currently_allowed or role not in ('student','teacher'):
            return {'status':'denied'}
        if not finite_nonnegative(now) or not finite_nonnegative(ttl) or ttl<=0:
            raise ValueError('finite nonnegative time and positive TTL required')
        key=cache_key(query,role,corpus_version,procedure_version)
        entry=self.entries.get(key)
```

```
        if entry is None:return {'status':'miss','reason':'no matching context key'}
        age=now-entry['saved_at']
        if age<0:return {'status':'miss','reason':'invalid negative age'}
        if age>=ttl:return {'status':'miss','reason':'expired'}
        return {'status':'hit','age_minutes':age,'value':deepcopy(entry['value'])}

cache=FixtureCache();query='room information';ttl=5
cache.put(query,'teacher','c1','p1',{'answer':'fictional teacher-only detail'},0)
teacher_hit=cache.get(query,'teacher','c1','p1',4,ttl,currently_allowed=True)
student_miss=cache.get(query,'student','c1','p1',4,ttl,currently_allowed=True)
revoked=cache.get(query,'teacher','c1','p1',4,ttl,currently_allowed=False)
expired=cache.get(query,'teacher','c1','p1',5,ttl,currently_allowed=True)
assert teacher_hit['status']=='hit' and student_miss['status']=='miss'
assert revoked=={'status':'denied'} and expired=={'status':'miss','reason':'expired'}
assert cache.get(query,'teacher','c2','p1',1,ttl,currently_allowed=True)['status']=='miss'
assert cache.get(query,'teacher','c1','p2',1,ttl,currently_allowed=True)['status']=='miss'
print('Main role/current-permission/expiry
results:',teacher_hit['status'],student_miss,revoked,expired)
# Separate pictured source versions; trusted publication context chooses current v5.
openings={'4':'09:00','5':'09:30'};current_version='5';lookups=[]
picture=FixtureCache();room='map-room opening'
picture.put(room,'student','4','p1',{'opening':'09:00','source_version':'4'},0)
first=picture.get(room,'student',current_version,'p1',1,ttl,currently_allowed=True)
assert first['status']=='miss'
lookups.append(current_version)  # local source fixture lookup, not a network call
picture.put(room,'student',current_version,'p1',
            {'opening':openings[current_version],'source_version':current_version},1)
next_request=picture.get(room,'student',current_version,'p1',2,ttl,currently_allowed=True)
assert next_request['status']=='hit' and next_request['value']['opening']=='09:30' and
lookups==['5']
print('Picture current-version first miss:',first)
print('Picture later matching fresh hit:',next_request,'source fixture lookups:',lookups)
# Deliberate hypothetical publication fault: same version label, changed source value.
silent_edit={'5':'09:45'}
assert next_request['value']['opening']!=silent_edit['5']
print('A silent unversioned edit would not be detected by this matching key/TTL alone.')
print('Expired/mismatched entries may remain in memory; serving validity is not proof of
retention deletion.')
print('Roles, permissions, approved versions and cached authorized values are supplied
fixtures; no real authentication, clock, provider or distributed cache.')
```

Expected outcomes and limits

1. Predict each main status before running or tracing: teacher hit, student miss, revoked teacher denied, exact-age-5 expired miss, and changed corpus/procedure misses. In the separate opening-time picture, a version 5 request first misses, reads one local approved fixture, then a later request hits 09:30. The hypothetical silent edit demonstrates a publication limitation rather than a repaired cache.

   Values are copied on save and return. The code supplies trusted roles and permissions; it implements no real login, network access, clock, distributed invalidation or retention deletion. Keep “not run” if you only trace it.

Evidence ledger, trace instructions and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

```
def cache_key(query, role, corpus_version, procedure_version):
    return (query, role, corpus_version, procedure_version)

def cache_valid(age, ttl, saved_version, current_version):
    return 0 <= age < ttl and saved_version == current_version

print(cache_valid(10, 10, "v1", "v1"))
```

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: combine a new TTL with a source-version change

> Separate the reason age 9 fails from the expiry at ages 10 and 11.

Complete selected independent task

TTL 10 minutes; ages 9, 10, 11. Corpus version changes at age 9. Decide validity.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Attempt this without hints

TTL 10 minutes; ages 9, 10, 11. Corpus version changes at age 9. Decide validity.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

</details>

MAIN LESSON · SECTION 11

Save context versions, timestamps and decision evidence

> Use fictional values rather than private cached content in the classroom record.

Project task and artifact

Add a labelled caching and stale information **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of caching and stale information.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain why fresh does not always mean reusable

> Name current permission, context and time before concluding.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of caching and stale information

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Cache key	Fields identifying a reusable result
TTL	The lifetime allowed by the cache contract
Invalidation	Marking an entry unusable after a relevant change

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Caching and stale information — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Teacher response is cached by query only and returned to a student.	Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.	Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

By the end, I can	How I will show it
Explain the mechanism and terminology of caching and stale information	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.
- Which labelled step matches this detail? Retrieve and cache version 5’s 09:30 under its own key.
- 1. Read authority
- 2. Fetch current **evidence**
- 3. Inspect cached key
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Fetch current **evidence**
Retrieve and cache version 5’s 09:30 under its own key. A cache saves repeated work but can retain obsolete facts.

Version-aware keys distinguish answers grounded in different records.

Changed-case question and optional hint

- Changed-case practice: The next request still uses approved version 5. Which conclusion is supported by the supplied facts?
- 1. The matching version 5 entry can supply 09:30 under the freshness policy.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The matching version 5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

Repair and retain the record

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

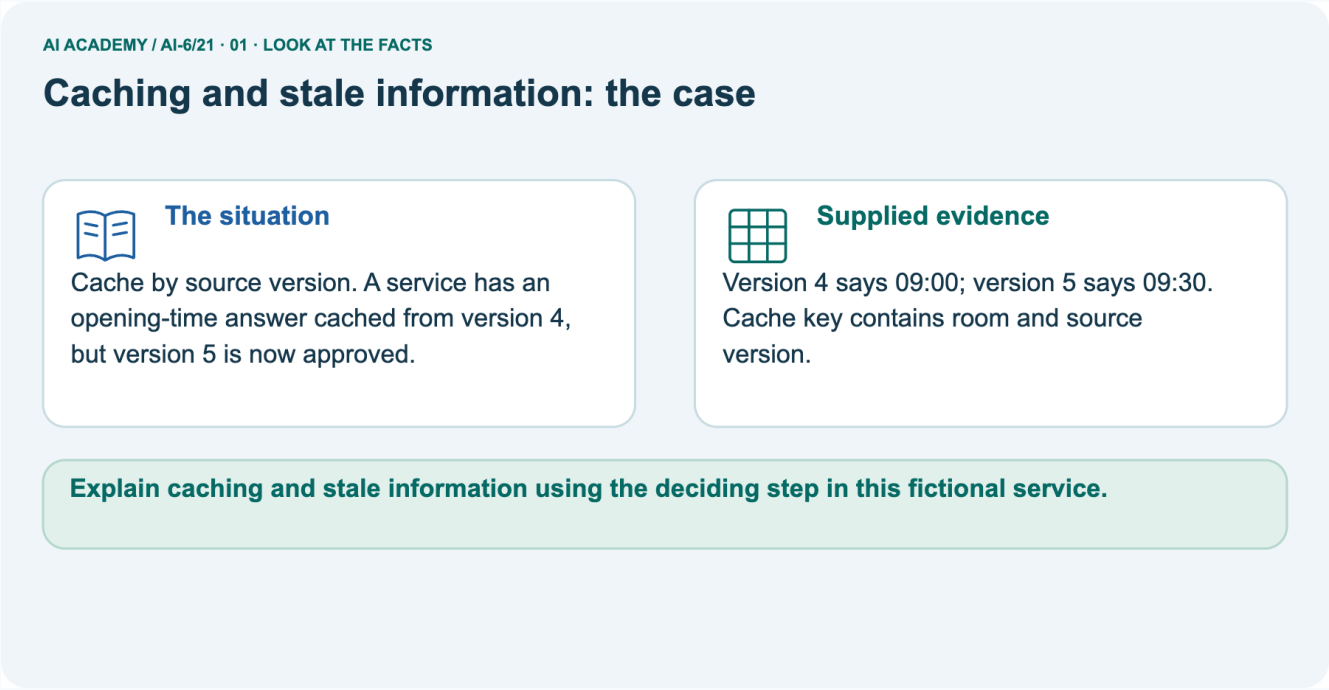
Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Caching and stale information: the case

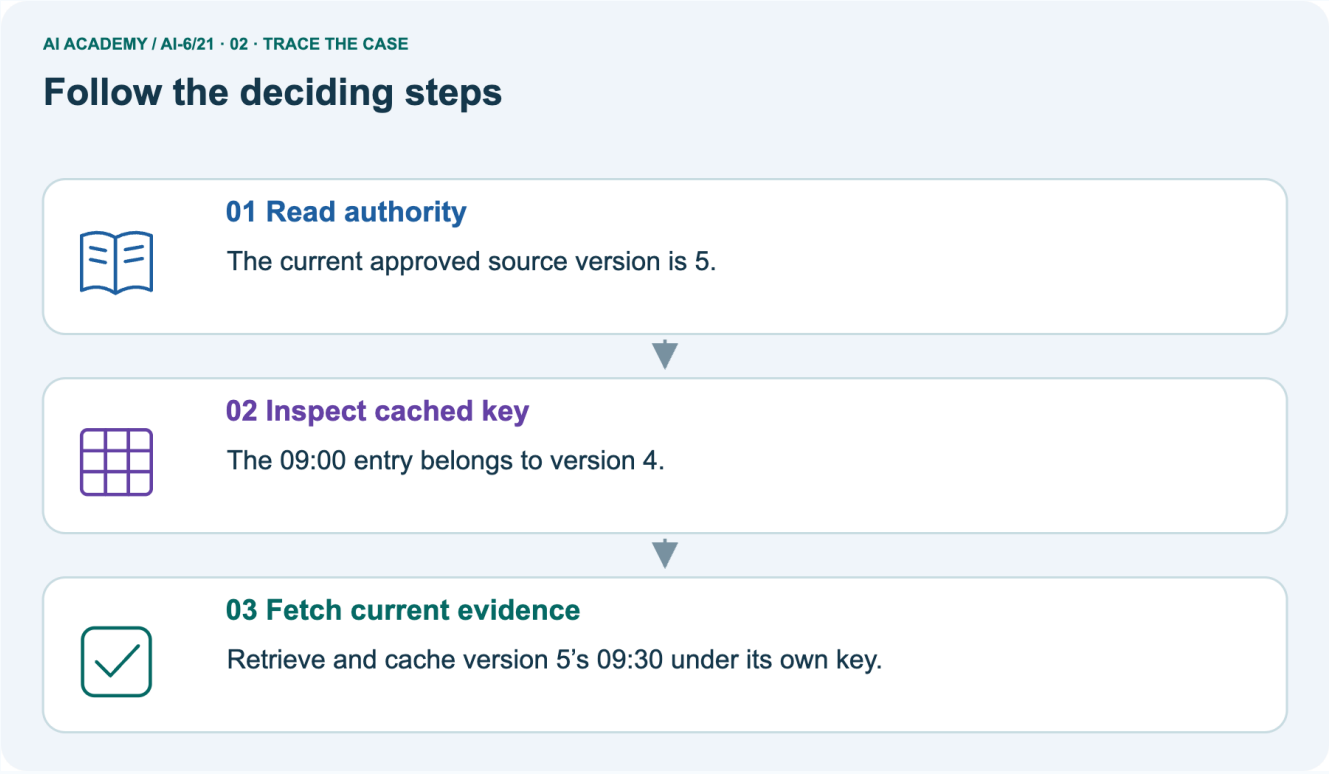


Follow the labelled picture with the matching explanation. Open the vector diagram

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved. Supplied facts: Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

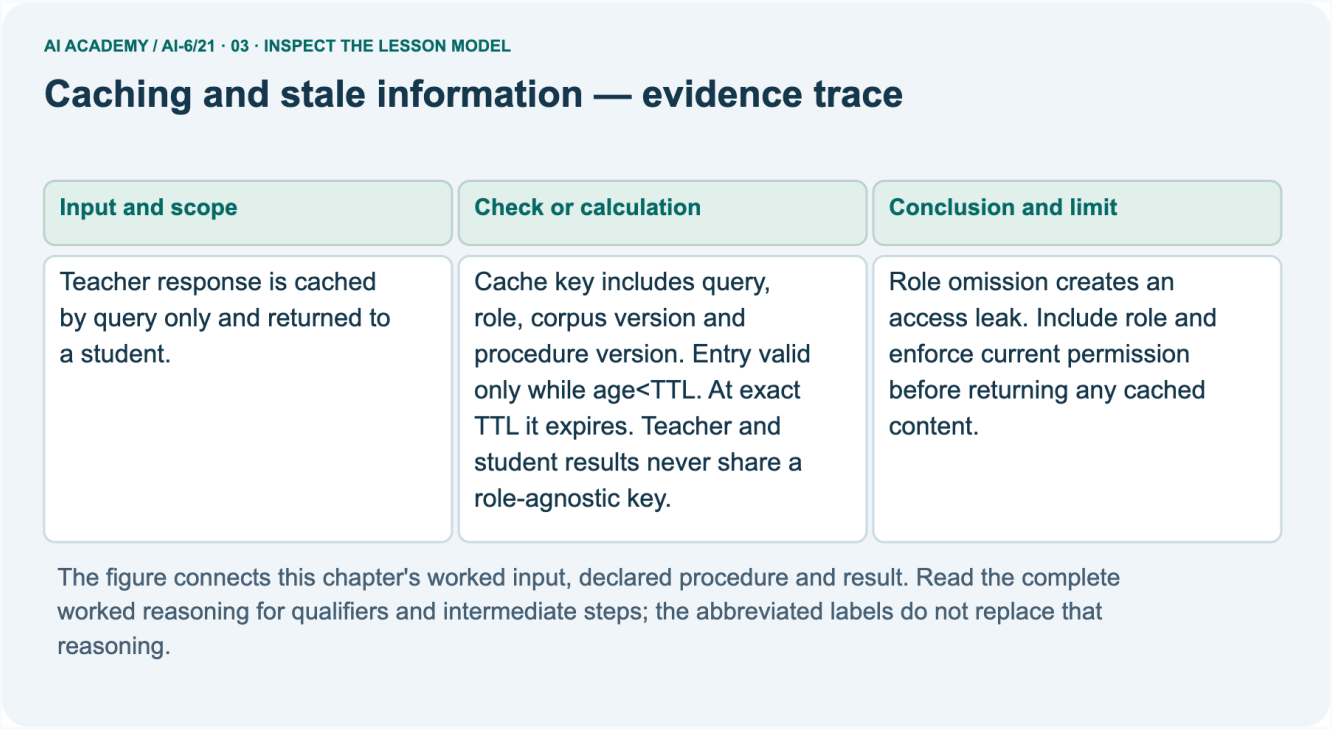


Follow the labelled picture with the matching explanation. Open the vector diagram

Read authority: The current approved source version is 5. → Inspect cached key: The 09:00 entry belongs to version 4. → Fetch current **evidence**: Retrieve and cache version 5's 09:30 under its own key.

Point to each numbered stage and explain which recorded **input** it uses.

03. Caching and stale information — evidence trace



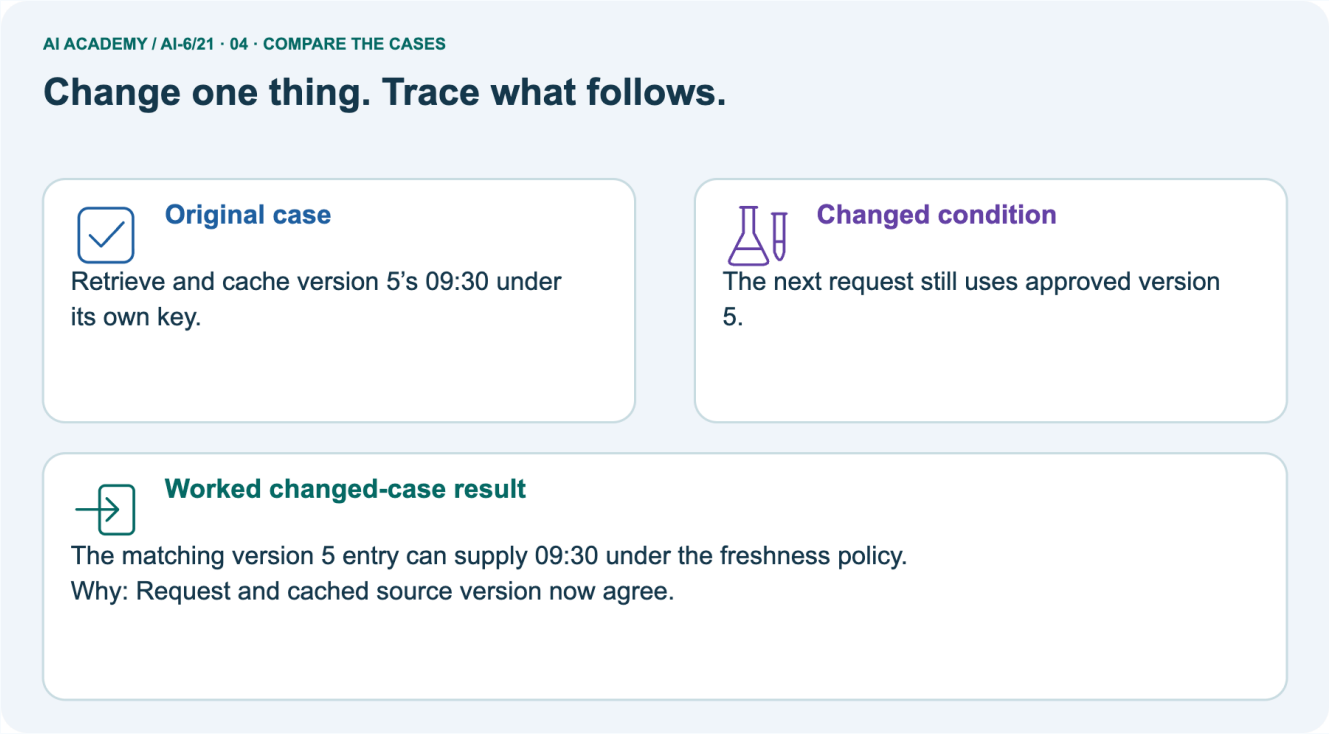
Follow the labelled picture with the matching explanation. Open the vector diagram

Caching and stale information — **evidence** trace. Teacher response is cached by query only and returned to a student. | **Cache key** includes query, role, corpus version and procedure version. Entry valid only while age<TTL.

At exact TTL it expires. Teacher and student results never share a role-agnostic key. | Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

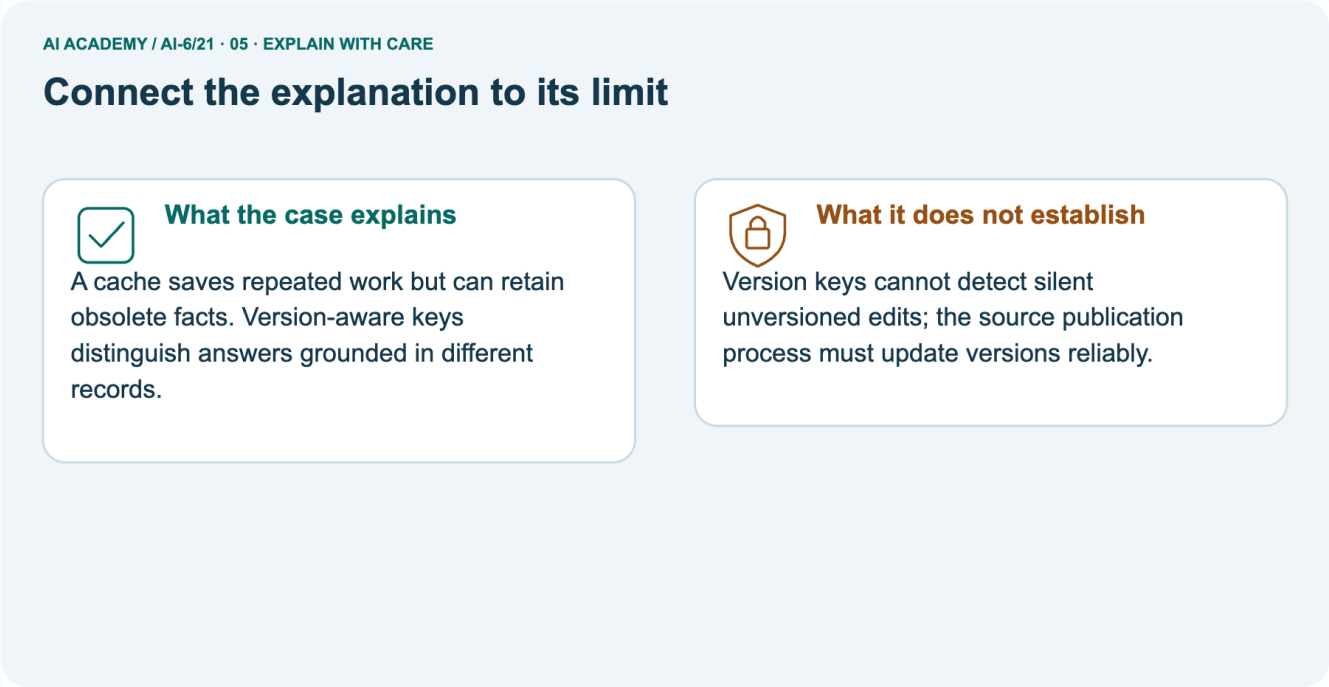


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retrieve and cache version 5's 09:30 under its own key. Changed condition: The next request still uses approved version 5. Expected result: The matching version 5 entry can supply 09:30 under the freshness policy. Reason: Request and cached source version now agree.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



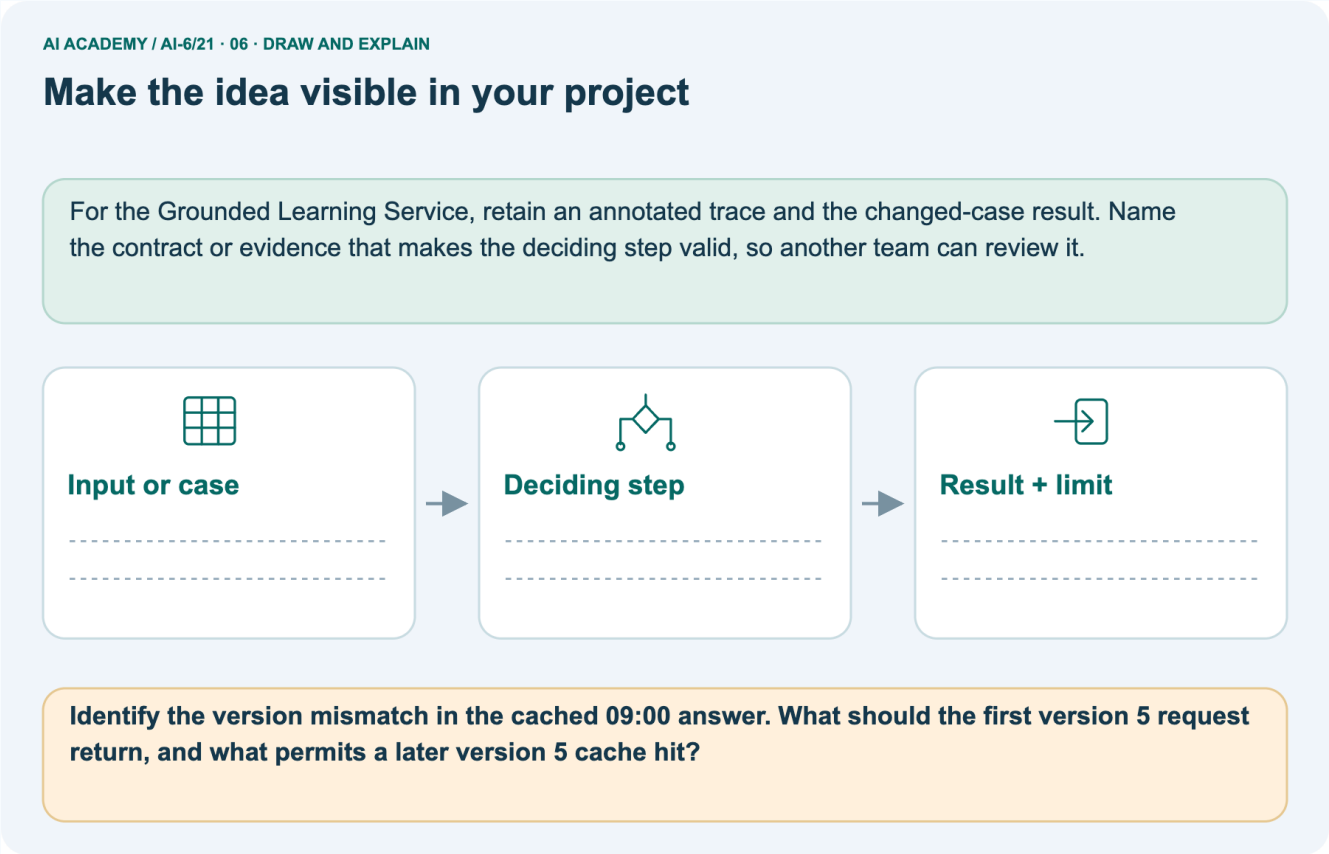
Follow the labelled picture with the matching explanation. Open the vector diagram

A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

Limit: Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it. Draw your **input**, deciding step, result and limit. Supported question: Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Caching and stale information: the case

02. Follow the deciding steps

03. Caching and stale information — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.